CSCI E-106:Assignment 1

```r
library(faraway)
data(prostate, package="faraway")

summary(prostate)
```

```
##      lcavol           lweight           age             lbph
##  Min.   :-1.3471   Min.   :2.375   Min.   :41.00   Min.   :-1.3863
##  1st Qu.: 0.5128   1st Qu.:3.376   1st Qu.:60.00   1st Qu.:-1.3863
##  Median : 1.4469   Median :3.623   Median :65.00   Median : 0.3001
##  Mean   : 1.3500   Mean   :3.653   Mean   :63.87   Mean   : 0.1004
##  3rd Qu.: 2.1270   3rd Qu.:3.878   3rd Qu.:68.00   3rd Qu.: 1.5581
##  Max.   : 3.8210   Max.   :6.108   Max.   :79.00   Max.   : 2.3263
##       svi              lcp            gleason          pgg45
##  Min.   :0.0000   Min.   :-1.3863   Min.   :6.000   Min.   :  0.00
##  1st Qu.:0.0000   1st Qu.:-1.3863   1st Qu.:6.000   1st Qu.:  0.00
##  Median :0.0000   Median :-0.7985   Median :7.000   Median : 15.00
##  Mean   :0.2165   Mean   :-0.1794   Mean   :6.753   Mean   : 24.38
##  3rd Qu.:0.0000   3rd Qu.: 1.1786   3rd Qu.:7.000   3rd Qu.: 40.00
##  Max.   :1.0000   Max.   : 2.9042   Max.   :9.000   Max.   :100.00
##       lpsa
##  Min.   :-0.4308
##  1st Qu.: 1.7317
##  Median : 2.5915
##  Mean   : 2.4784
##  3rd Qu.: 3.0564
##  Max.   : 5.5829
```

a-) How many observations are in the dataset?

```r
rows = nrow(prostate)
```

There are 97 observations in the dataset.

b-) Calculate the mean and median of each variable. Discuss whether any variables appear to have outliers, and briefly describe how you identified them.(20 points)

```r
variables = names(prostate)
response_variable = "lpsa"
explanatory_variables = variables[variables != response_variable]

stats = data.frame(
  Mean = sapply(prostate, mean, na.rm = TRUE),
  Median = sapply(prostate, median, na.rm = TRUE),
  StandardDeviation = sapply(prostate, sd, na.rm = TRUE)
)
print(stats)
```

```
##                Mean    Median StandardDeviation
## lcavol    1.3500096  1.446919         1.1786249
```

```
## lweight  3.6526887  3.623000          0.4966286
## age     63.8659794 65.000000          7.4451171
## lbph     0.1003558  0.300105          1.4508065
## svi      0.2164948  0.000000          0.4139949
## lcp     -0.1793637 -0.798510          1.3982478
## gleason  6.7525773  7.000000          0.7221341
## pgg45   24.3814433 15.000000         28.2040346
## lpsa     2.4783870  2.591520          1.1543287
```

A typical test for outliers selects points more than 1.5*IQR beyond the 1st/3rd quartiles.

```r
stats$Q1 = sapply(prostate, quantile, na.rm = TRUE)["25%",]
stats$Q3 = sapply(prostate, quantile, na.rm = TRUE)["75%",]
stats$IQR = stats[,"Q3"] - stats[,"Q1"]
```

```r
stats$LowerBound = stats[,"Q1"] - (1.5*stats[,"IQR"])
stats$UpperBound = stats[,"Q3"] + (1.5*stats[,"IQR"])

stats$Outliers = lapply(variables,
                        function(n) {
                          length(
                            prostate[,n][prostate[,n] < stats["LowerBound",n]]
                            | prostate[,n][prostate[,n] < stats["UpperBound",n]]
                            ) })
for (variable in variables) {

}
print(stats)
```

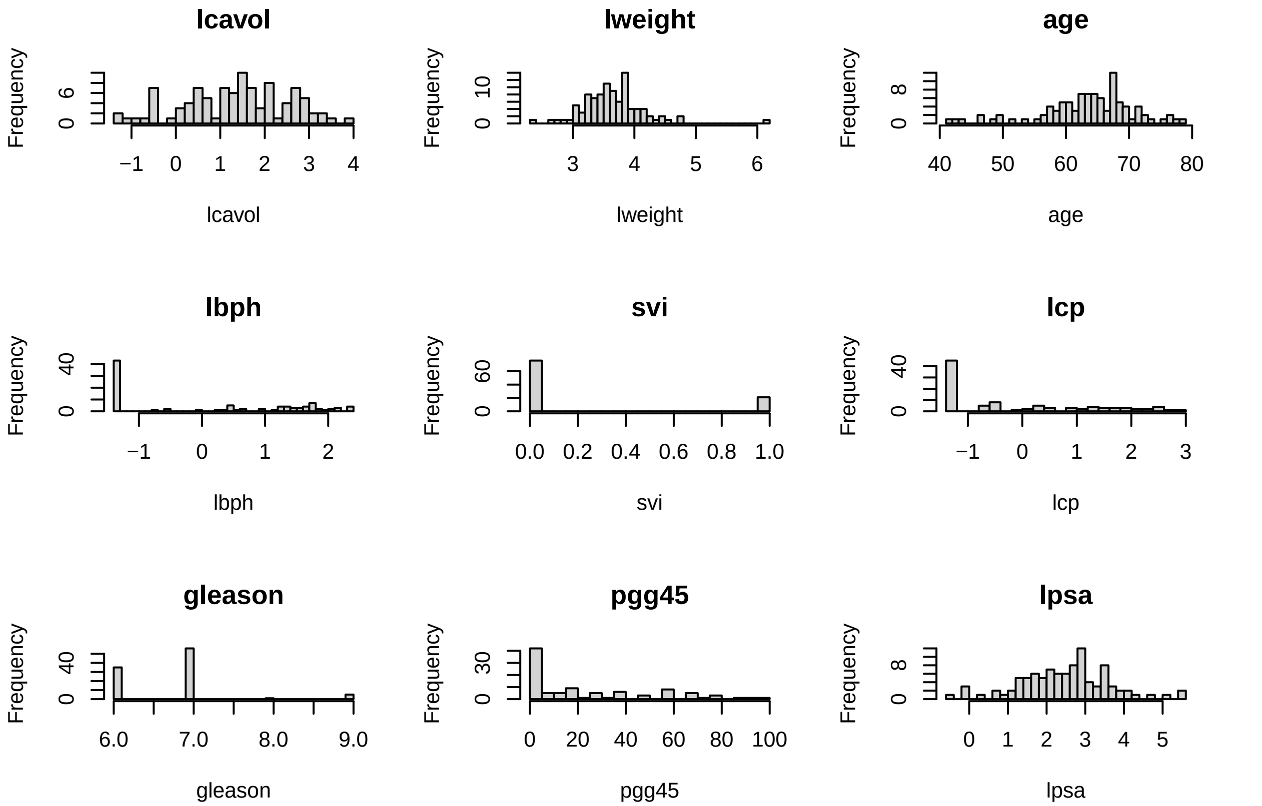
```
##               Mean    Median StandardDeviation         Q1        Q3       IQR
## lcavol   1.3500096  1.446919         1.1786249  0.5128236  2.127041  1.614217
## lweight  3.6526887  3.623000         0.4966286  3.3759000  3.878500  0.502600
## age     63.8659794 65.000000         7.4451171 60.0000000 68.000000  8.000000
## lbph     0.1003558  0.300105         1.4508065 -1.3862940  1.558145  2.944439
## svi      0.2164948  0.000000         0.4139949  0.0000000  0.000000  0.000000
## lcp     -0.1793637 -0.798510         1.3982478 -1.3862900  1.178650  2.564940
## gleason  6.7525773  7.000000         0.7221341  6.0000000  7.000000  1.000000
## pgg45   24.3814433 15.000000        28.2040346  0.0000000 40.000000 40.000000
## lpsa     2.4783870  2.591520         1.1543287  1.7316600  3.056360  1.324700
##         LowerBound UpperBound Outliers
## lcavol   -1.908502   4.548366        0
## lweight   2.622000   4.632400        0
## age      48.000000  80.000000        0
## lbph     -5.802952   5.974804        0
## svi       0.000000   0.000000        0
## lcp      -5.233700   5.026060        0
## gleason   4.500000   8.500000        0
## pgg45   -60.000000 100.000000        0
## lpsa     -0.255390   5.043410        0
```

The analysis does not identify any outliers.

The pgg45 variable has a lot of 0 values, which may indicate a coding that used 0 instead of n/a.

The gleason variable is strongly centered around the mean of 6.753, with a standard deviation of 0.722, but about 5% of observations are 8s and 9s.

```r
par(mfrow = c(ceiling(ncol(prostate)/3), 3))
for(col in variables) {
  hist(prostate[[col]], main = col,
       xlab = col, breaks = 30)
}
```



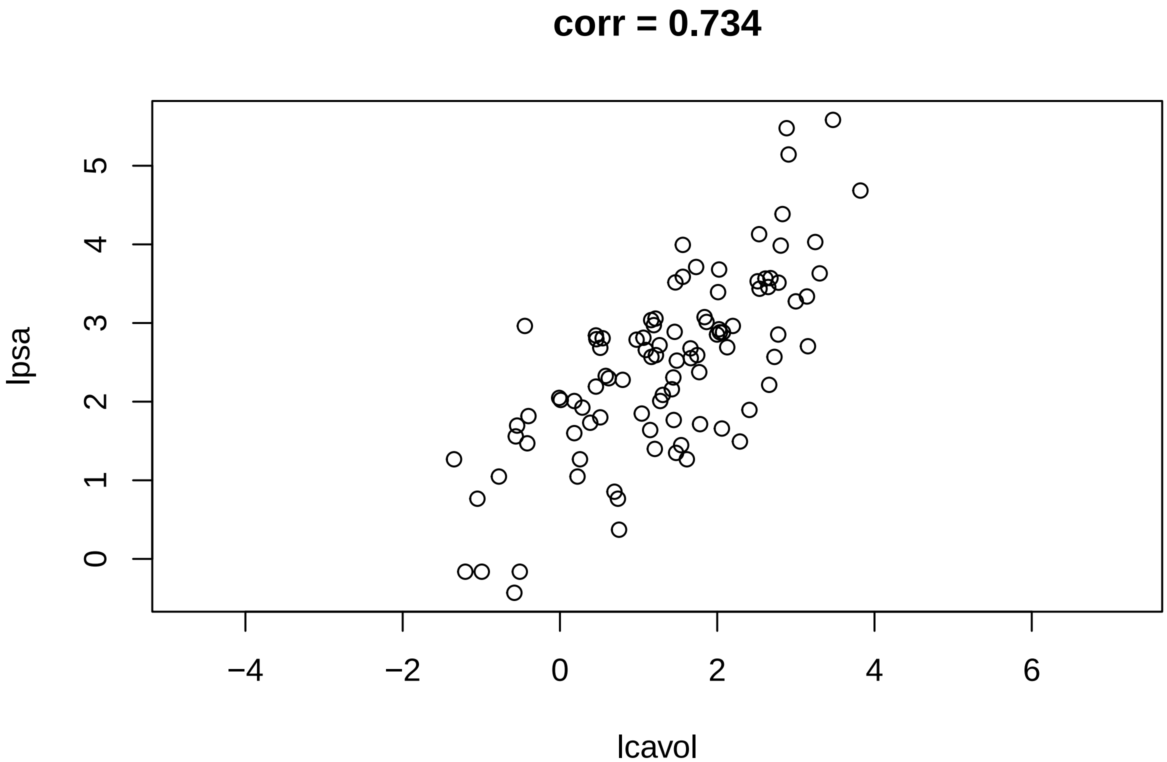
c-) Compute the correlation matrix for all variables.Create scatterplots of each predictor versus the response variable lpsa. Identify which predictors appear to be strongly correlated with lpsa, using both the correlations and the plots to justify your answer. (20 points)

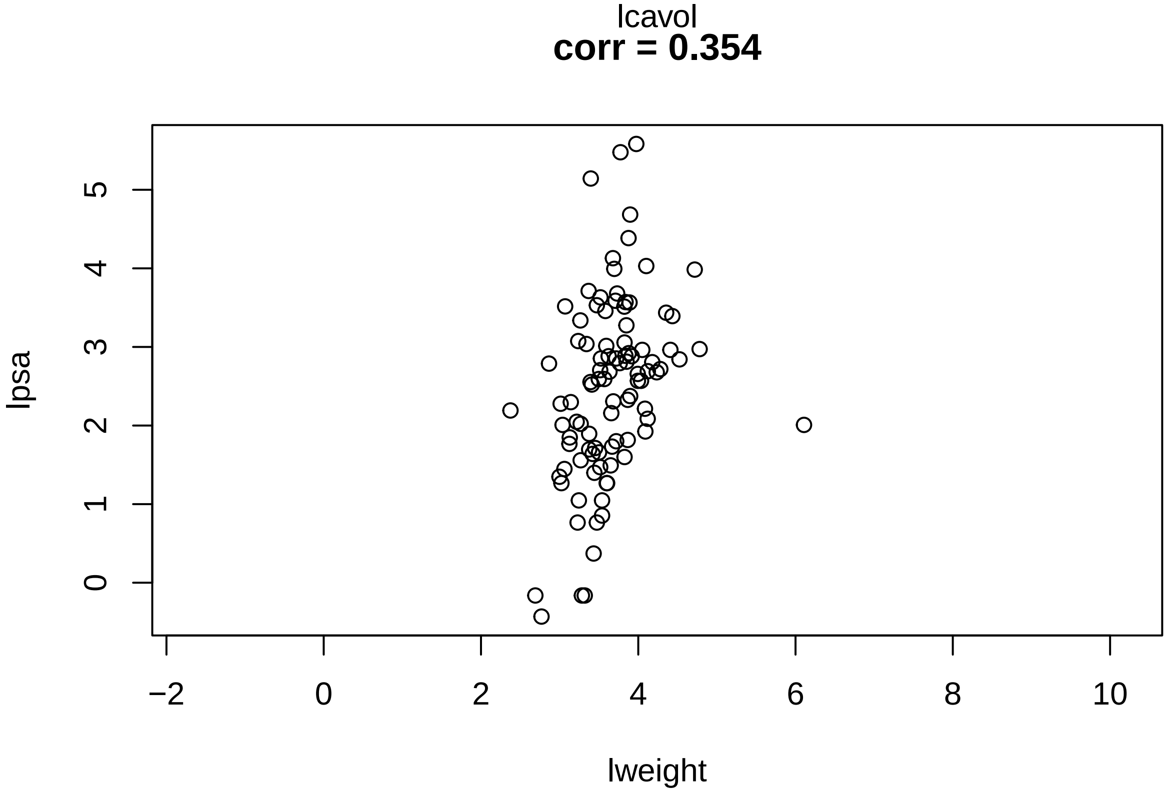
```r
correlations <- cor(prostate)
print(round(correlations, 2))
```

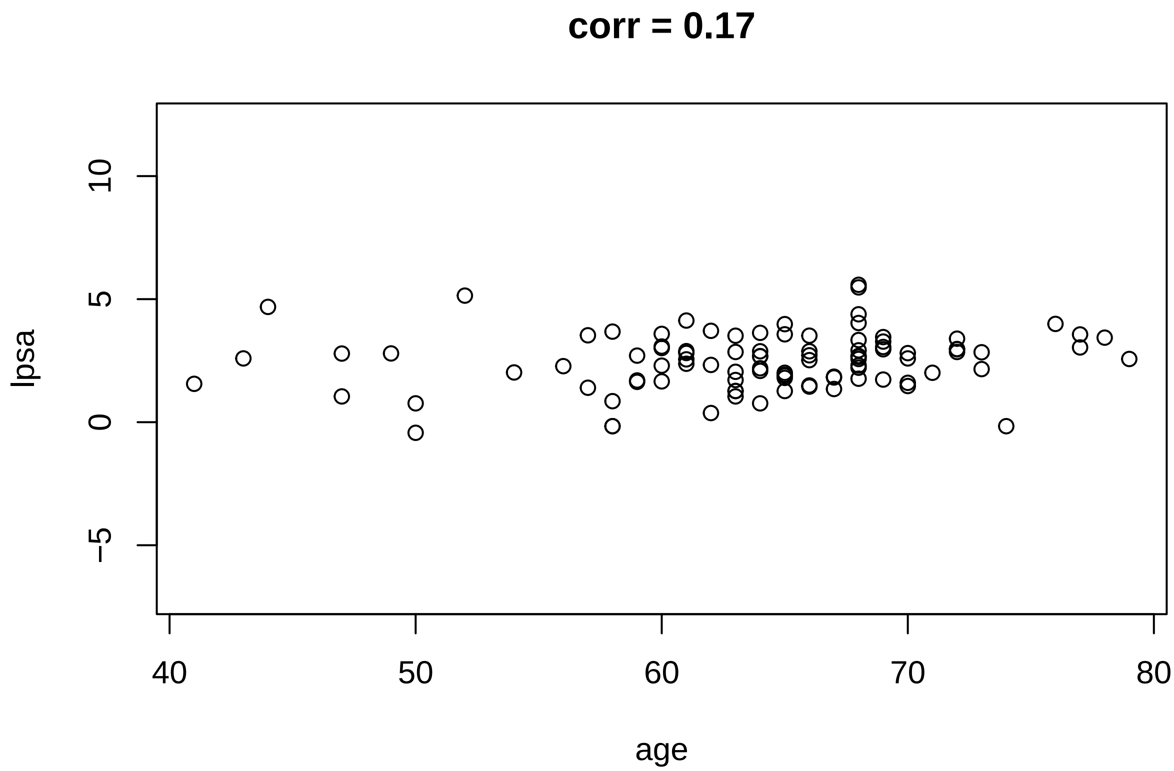
```
##         lcavol lweight  age  lbph   svi   lcp gleason pgg45 lpsa
## lcavol    1.00    0.19 0.22  0.03  0.54  0.68    0.43  0.43 0.73
## lweight   0.19    1.00 0.31  0.43  0.11  0.10    0.00  0.05 0.35
## age       0.22    0.31 1.00  0.35  0.12  0.13    0.27  0.28 0.17
## lbph      0.03    0.43 0.35  1.00 -0.09 -0.01    0.08  0.08 0.18
## svi       0.54    0.11 0.12 -0.09  1.00  0.67    0.32  0.46 0.57
## lcp       0.68    0.10 0.13 -0.01  0.67  1.00    0.51  0.63 0.55
## gleason   0.43    0.00 0.27  0.08  0.32  0.51    1.00  0.75 0.37
## pgg45     0.43    0.05 0.28  0.08  0.46  0.63    0.75  1.00 0.42
## lpsa      0.73    0.35 0.17  0.18  0.57  0.55    0.37  0.42 1.00
```

```r
par(mfrow=c(1,1))
for(name in explanatory_variables) {
  plot(prostate[,name], prostate[,response_variable],
       asp=1,
       xlab = name,
```

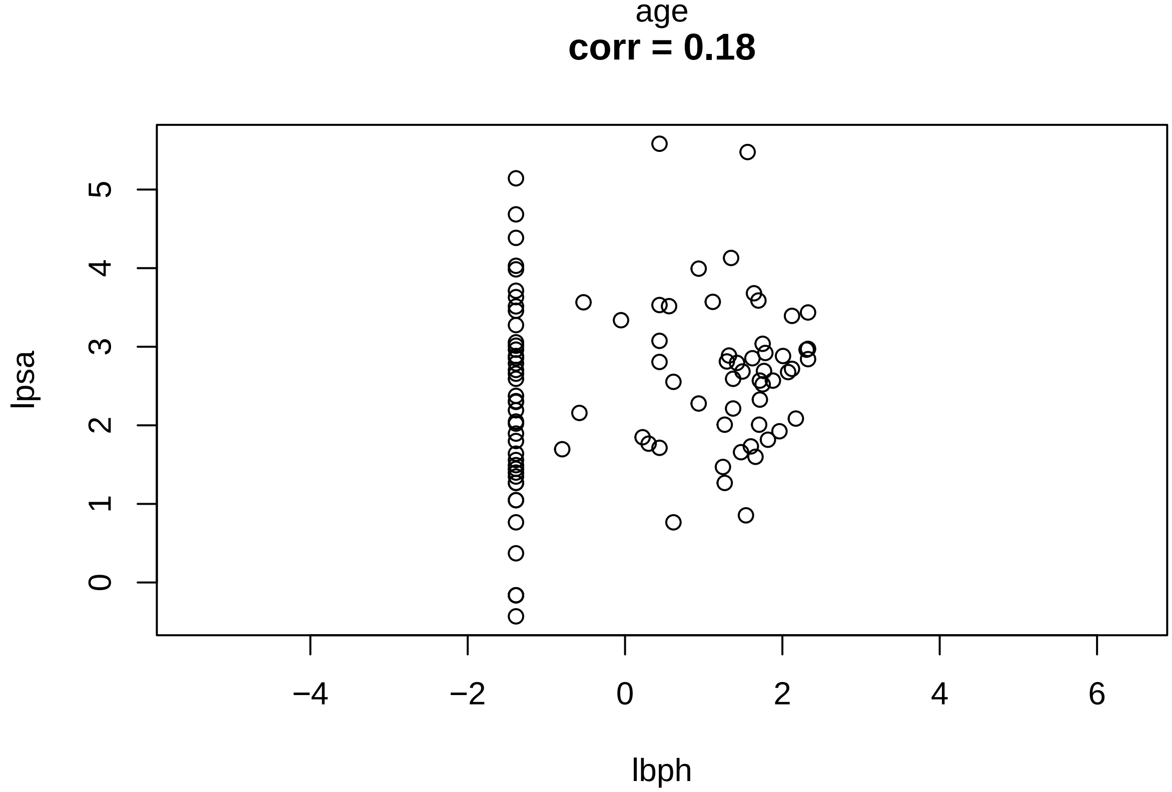
```
        ylab = response_variable,
        main=paste("corr =", round(correlations[name, response_variable], 3)))
}
```



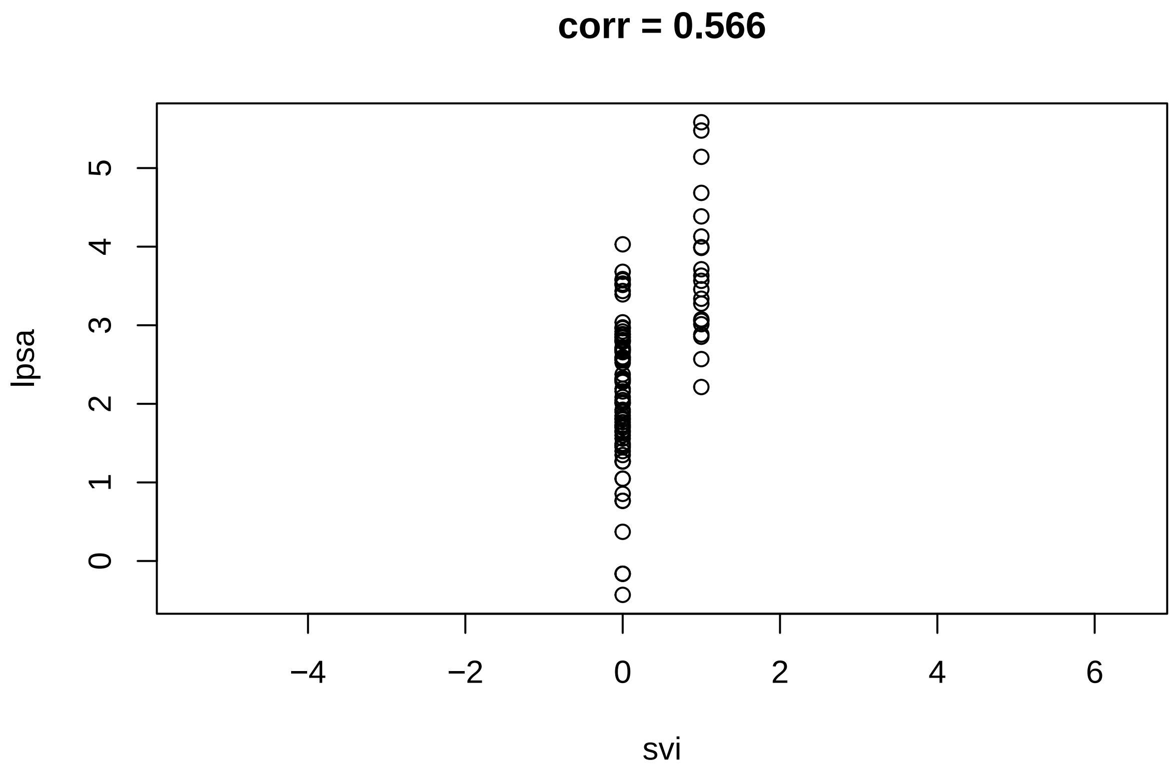


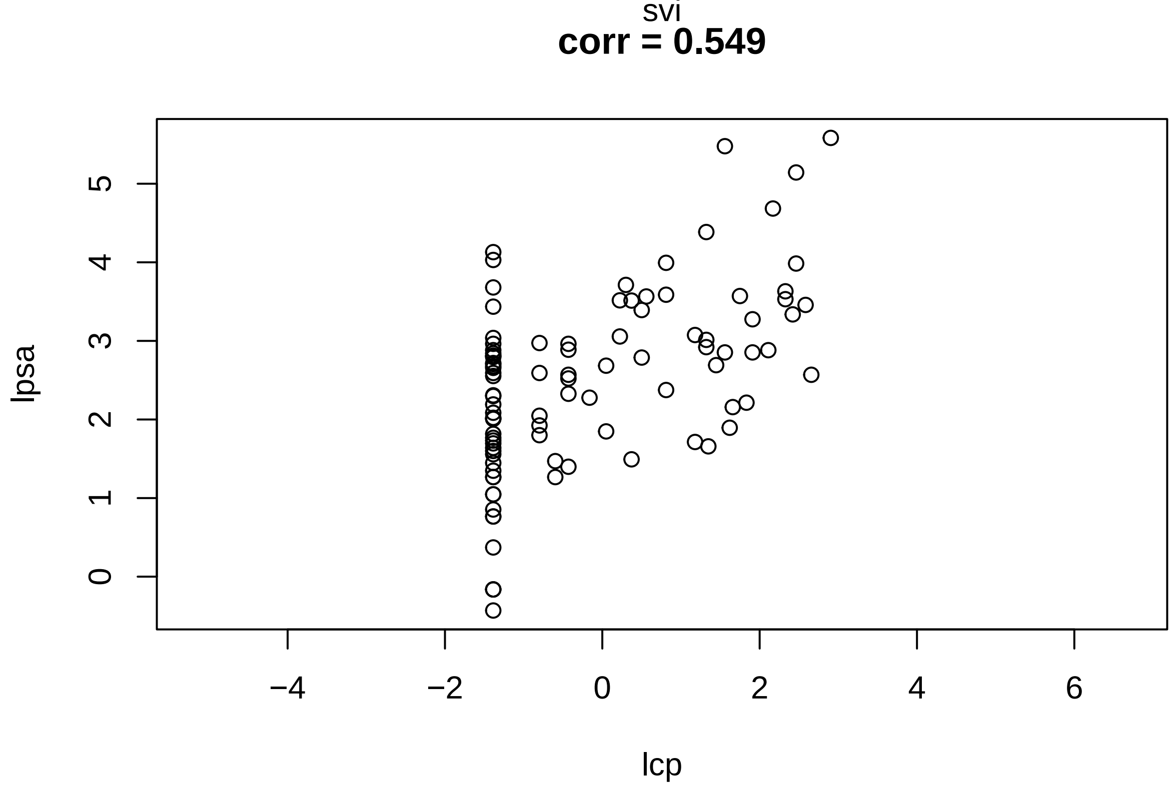


corr = 0.17
lpsa
10
5
0
−5
40
50
60
70
80
age

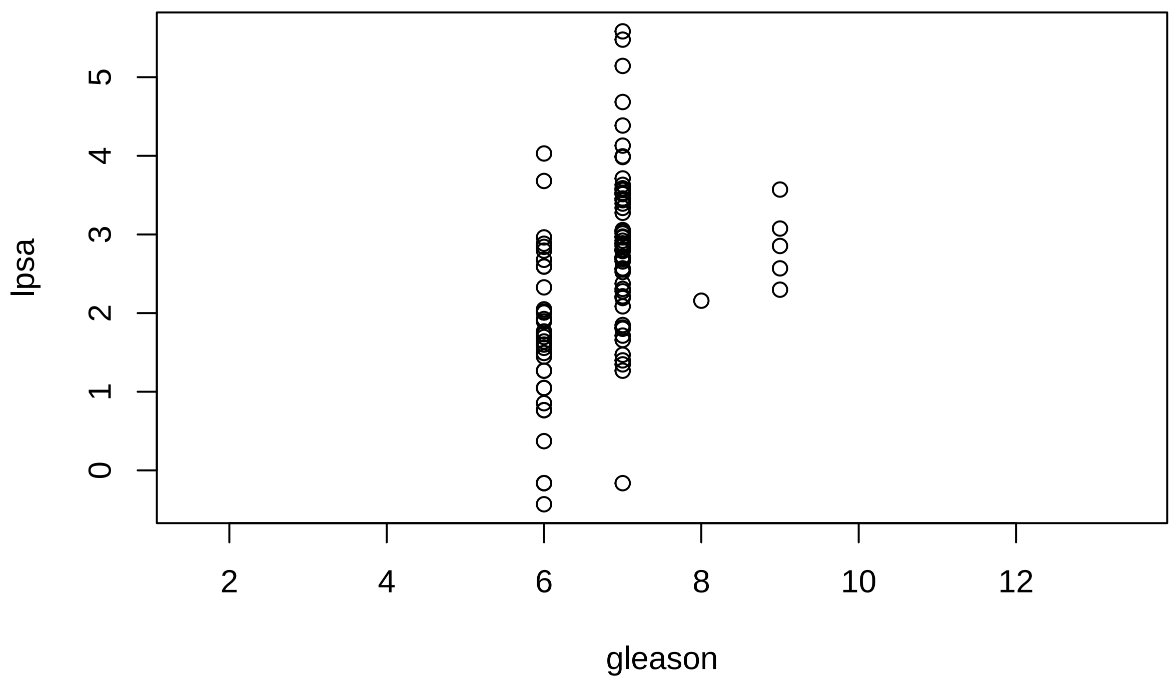


corr = 0.18
lpsa
5
4
3
2
1
0
−4
−2
0
2
4
6
lbph

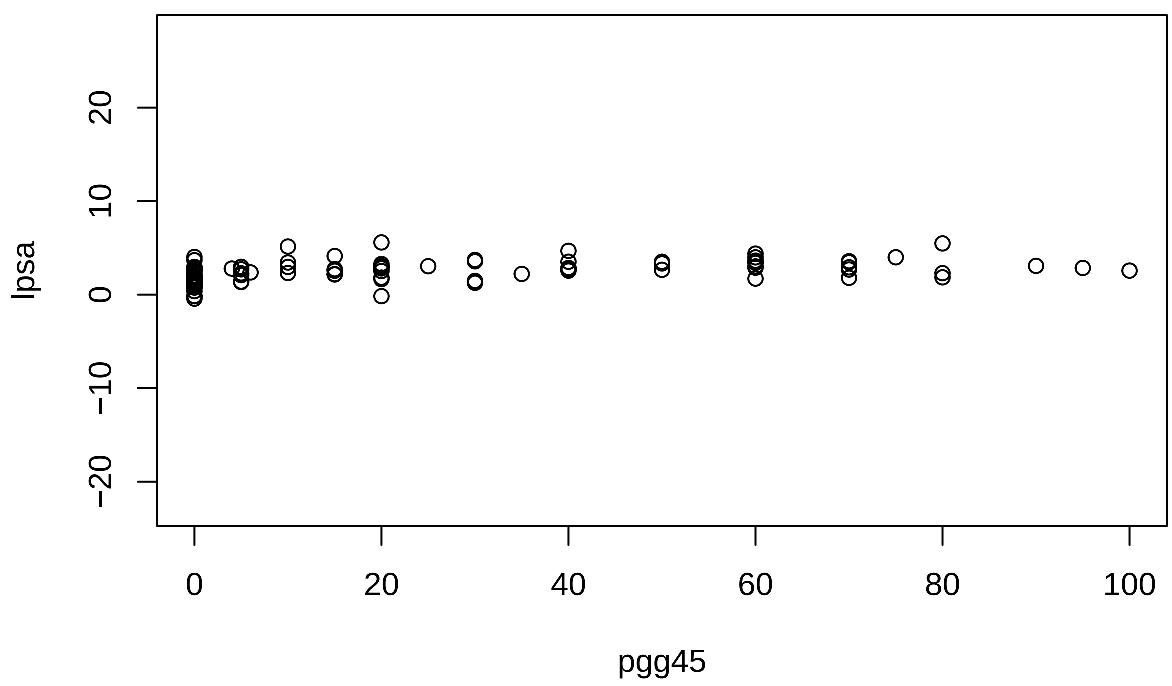

corr = 0.566
lpsa
svi
−4
−2
0
2
4
6
0
1
2
3
4
5


corr = 0.549
lpsa
lcp
−4
−2
0
2
4
6
0
1
2
3
4
5

corr = 0.369



corr = 0.422



TODO: Interpret

d-) Build separate simple linear regression models, by using lcavol, lweight and age to explain lpsa. For each model, clearly state the fitted regression equation with estimated coefficients. (25 points)

```r
lm_lcavol = lm(lpsa ~ lcavol, prostate)
print(lm_lcavol)
```

```
##
## Call:
## lm(formula = lpsa ~ lcavol, data = prostate)
##
## Coefficients:
## (Intercept)       lcavol
##      1.5073       0.7193
```

```r
lm_lweight = lm(lpsa ~ lweight, prostate)
print(lm_lweight)
```

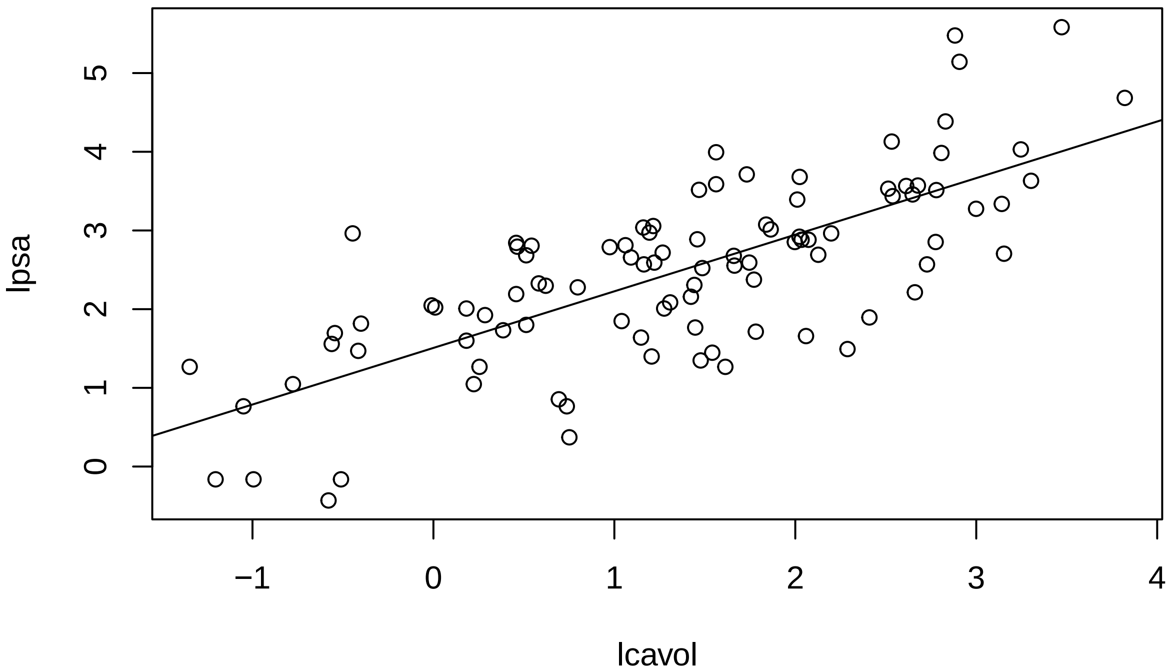
```
##
## Call:
## lm(formula = lpsa ~ lweight, data = prostate)
##
## Coefficients:
## (Intercept)      lweight
##     -0.5281       0.8231
```

```r
lm_age = lm(lpsa ~ age, prostate)
print(lm_age)
```

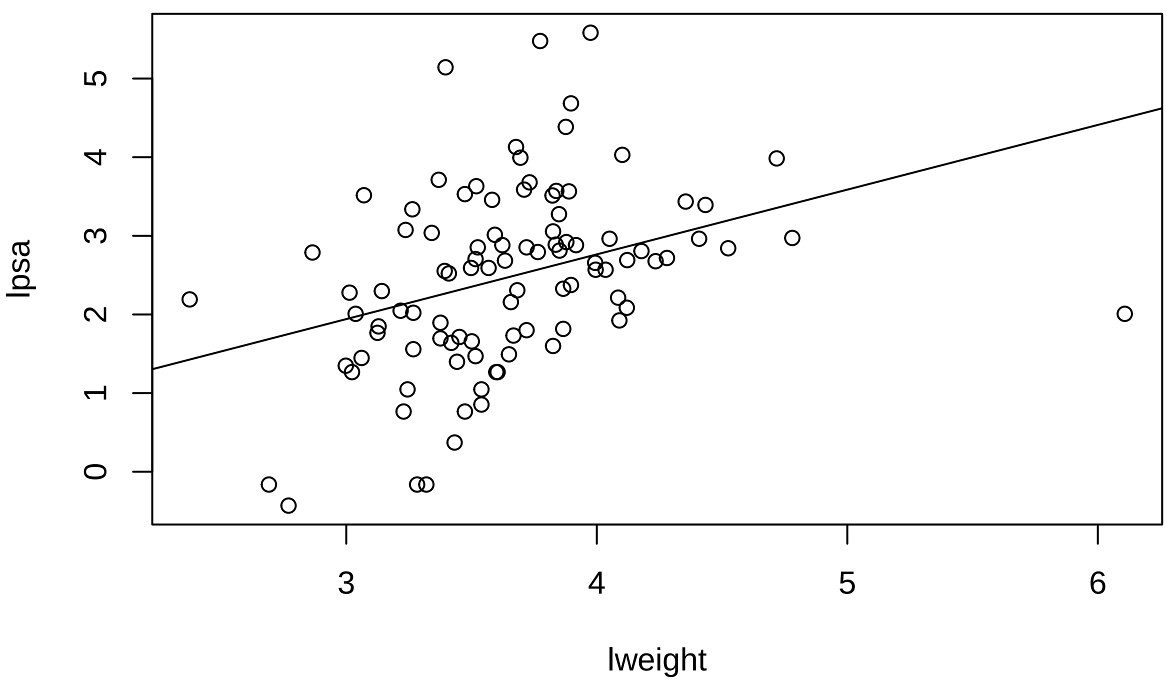
```
##
## Call:
## lm(formula = lpsa ~ age, data = prostate)
##
## Coefficients:
## (Intercept)          age
##     0.79906      0.02629
```

e-) For each regression model from part (d):
- Plot lpsa versus each predictor separately.
- Add the fitted regression line to each plot.
- Based on the models and plots, choose the best single-predictor model and explain your reasoning(

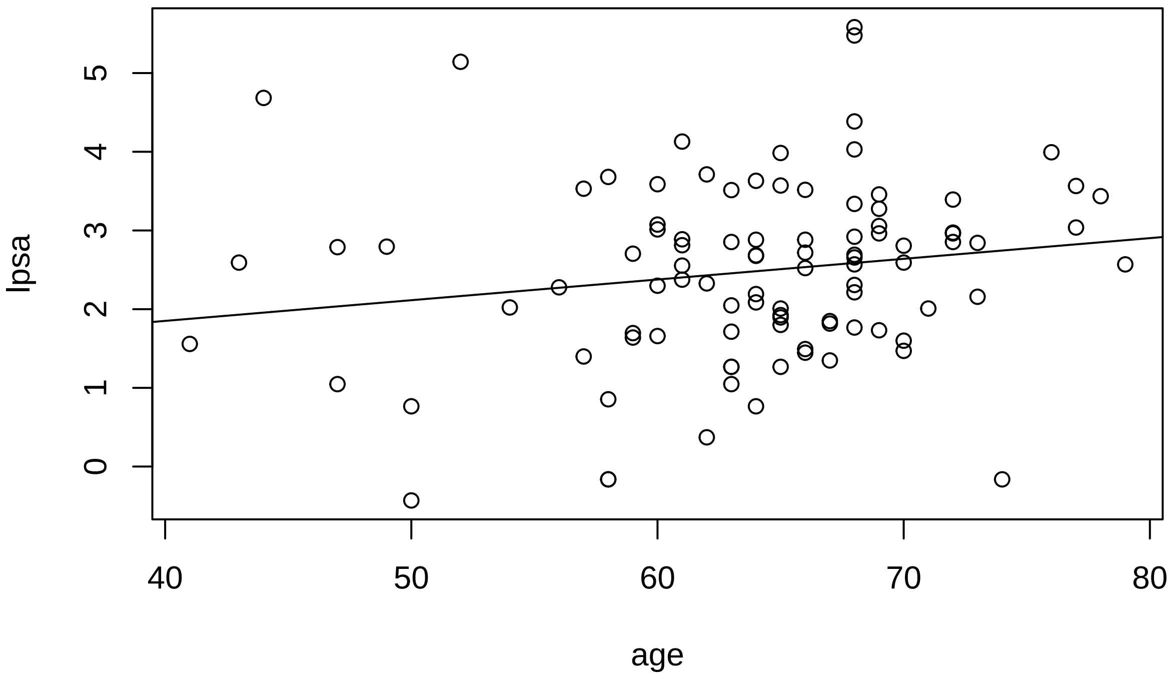
```r
plot(lpsa ~ lcavol, prostate)
abline(lm_lcavol)
```




```r
plot(lpsa ~ lweight, prostate)
abline(lm_lweight)
```




```r
plot(lpsa ~ age, prostate)
abline(lm_age)
```



TODO: Interp (c)